

Continuous Delivery Using Azure DevOps Services

CDADS | 2 Days



Scale your team's agility and deliver value continuously with Azure DevOps. This two-day course teaches the DevOps principles and hands-on practices that help multiple teams work as one, from increasing flow and sharing code to integrating, delivering, and learning continuously on a shared case study.

Who should take this course?

This course is for experienced software development professionals who want to adopt DevOps to achieve Continuous Integration, Continuous Delivery, continuous feedback, and continuous learning across a scaled, multi-team value stream. Attendees should be familiar with C#, Visual Studio, and Scrum, and have basic experience with Azure DevOps.

Course content

This course progressively builds your fluency in scaling DevOps across teams, with over 50 hands-on activities, on a common case study in a shared Azure DevOps environment, spanning flow, planning, code sharing, continuous integration, delivery, and learning.

1. INCREASING FLOW AT SCALE

- Increasing flow through a technical value stream
- Empirical process control
- Scaling Scrum with the Nexus framework
- Organizing teams to minimize dependencies

2. PLANNING AND EXECUTING AT SCALE

- Setting up a new product and planning at scale
- Refining the Product Backlog; defining "ready"
- Dependencies and cross-team refinement
- Planning and executing a Sprint, limiting WIP
- Definition of Done and reporting data

3. SHARING CODE

- Collective ownership
- Git primer and branching strategies
- Managing dependencies with Code Maps
- Sharing binaries with Azure Artifacts
- Practicing internal open source (inner source)

4. INTEGRATING CONTINUOUSLY

- Automated and unit testing in Visual Studio
- Build pipelines and running tests in the pipeline
- Code coverage and regression testing
- Continuous Integration and team practices
- Continuous Integration+

5. DELIVERING CONTINUOUSLY

- Release pipelines, stages, and deployments
- Deploying to Microsoft Azure
- Creating and deploying a release with triggers
- Practicing Continuous Delivery (CD)

6. EMPOWERING THE PRODUCT OWNER

- Build, measure, learn and Hypothesis-Driven Development
- Feature flags for release control
- Telemetry and A/B testing
- Exploratory testing and testing tours
- Identifying and paying off technical debt

7. LEARNING AND IMPROVING CONTINUOUSLY

- Working and learning as a team
- Patterns of effective collaboration
- Pairing, swarming, and mobbing
- Blameless retrospectives and agile metrics
- Communities of Practice and tribal knowledge

Course Designer

This course was designed by Richard Hundhausen, Microsoft's first Visual Studio ALM/DevOps MVP, a Professional Scrum Trainer, and co-creator of the Nexus framework, with decades helping teams build and ship software. To see other Azure DevOps and GitHub courses, visit www.accentient.com.